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THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER-2015

ELCTRICNETWORK ANALYSIS

Time:3 Hours **Maximum Marks:70**

Instructions : Answer five full questions, choosing at least TWO from each part.

PART-A

1. a. Distinguish between the following network elements. **4**

i) Linear and non-linear

ii) Active and passive

b. In the network shown in Fig. 1(b), find I and I_1 using star–delta transformation. **4**

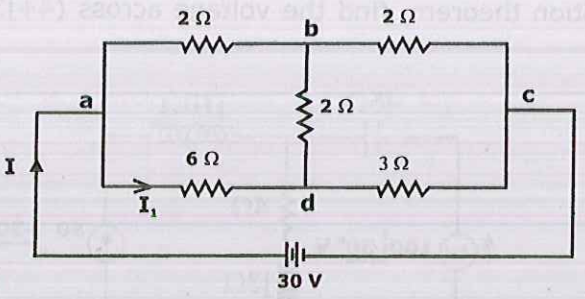

Fig1.(b)

c. Using mesh current analysis , find the current through 24Ω in the circuit shown in Fig. 1(c). **6**

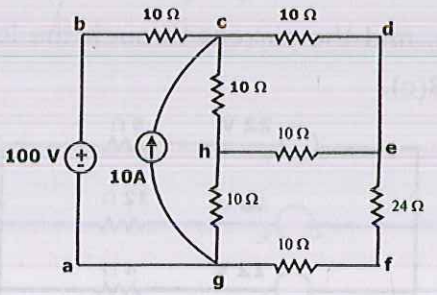

Fig. 1(c)

2. a. Define the following terms with reference to network topology. **4**

i) Graph ii) Branch iii) Node iv) Tree

b. Construct a tree for the network shown in Fig. 2(b), so that all loop currents pass through 7Ω. Write the corresponding Tie-set matrix. **6**

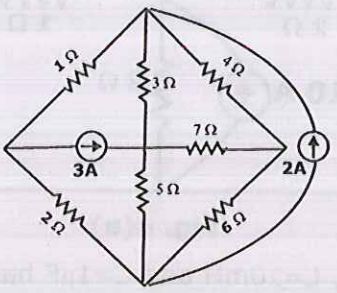

Fig. 2(b)

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c. Obtain the dual of the network graph shown in Fig. 2(c). **4**

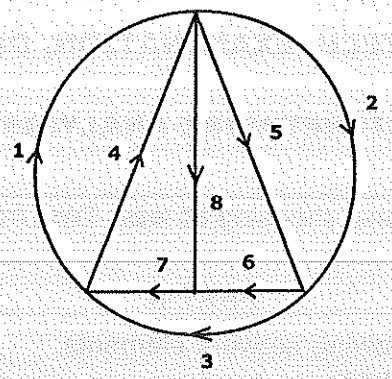

Fig. 2(c)

3. a. State and explain reciprocity theorem. **4**

b. Using superposition theorem, find the voltage across (4+j3)Ω in the circuit shown in Fig. 3(b). **5**

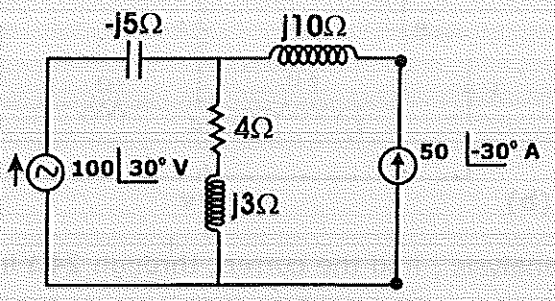

Fig. 3(b)

c. Using Millman's Theorem, find the current through the load resistance R_L=100Ω in the circuit shown in Fig. 3(c). **5**

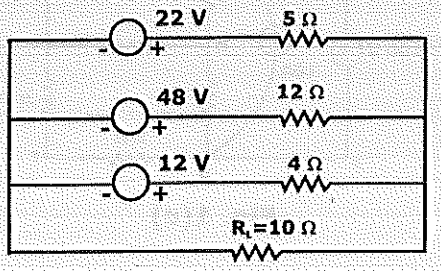

Fig. 3(c)

4. a. State the Thevenin's Theorem. For the circuit shown in Fig. 4(a), find the current through R_L using Thevenin's Theorem. **6**

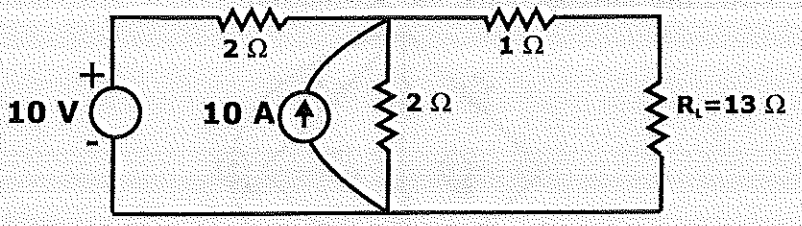

Fig. 4(a)

b. A series circuit with R=10Ω, L=10mH and C=1μF has an applied voltage of 200 V. Calculate the resonant frequency f_o, the current in the circuit at resonance and the voltage across the elements at resonance. Also find quality factor and bandwidth. **8**

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PART-B

5. a. A 3 phase delta connected load consumes a power of 60kw taking a lagging current of 200A at a line voltage of 400 V, 50 Hz. Find the parameters of each phase. What would be the power consumed, if the load were connected in star? **08**

b. A balanced 3phase star connected load draws power from 440 V supply. The two wattmeters connected indicate w_1=5 Kw and w_2=1.2Kw. Calculate total power, power factor and current in the circuit. **06**

6. a. In the circuit shown in Fig. 6(a), switch k is changed from position 1 to 2 at t=0. Steady state condition having reached before switching. Find i, $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0^+$. **06**

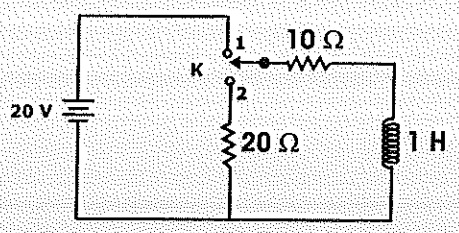

Fig. 6(a)

b. In the circuit shown in Fig. 6(b), find the current i(t) when the switch k is opened at t=0, with the circuit having reached steady state before switching. Find the current at t=0.5sec. **08**

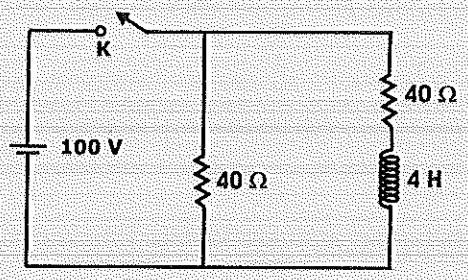

Fig. 6(b)

7. a. Assuming that the staircase waveform of Fig. 7(a) is not repeated, find its Laplace transform. If this voltage wave is applied to a R-L series circuit with R=1Ω and L= 1 H, find the current i(t). **06**

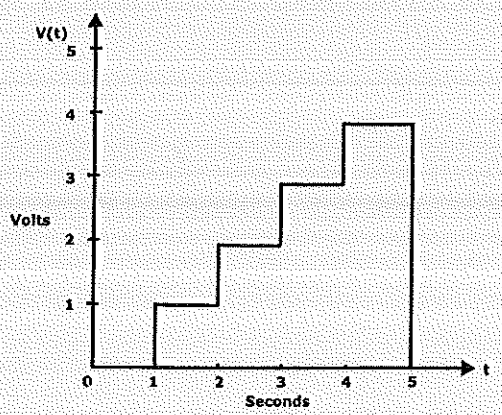

Fig. 7(a)

b. Find the initial and final values of the following functions using the theorems. **8**

i) $i(t)=3e^{-t}-e^{-2t}$

ii) $i(t)=5u(t)-3e^{-2t}$

8. a. Obtain the expression of z-parameters in terms of h-parameters. 6

b. The z-parameters of a two port network $z_{11}=20\Omega$, $z_{22}=30\Omega$, $z_{12}=z_{21}=10\Omega$. Find Y and ABCD parameters of the net work 8